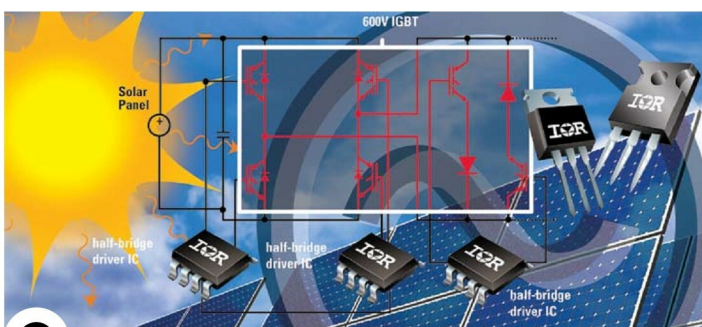




600 Volt Solar Inverter Solutions

IR's Energy Efficient Portfolio for Grid-Connected Solar Inverter Output Circuits

600V Fast Switching Trench IGBTs for PWM*

Part Number	Package Type	Rated Current $T_{CASE} = 100^{\circ}\text{C}$, $V_{GE} = 15\text{V}$	$V_{CE(on)}$	$t_d(on)$	t_r	$t_d(off)$	t_f
IRGB4059DPBF	TO-220	4.0A	2.15V	20 ns	15 ns	85 ns	35 ns
IRGB4045DPBF	TO-220	6.0A	2.14V	26 ns	12 ns	95 ns	32 ns
IRGB4060DPBF	TO-220	8.0A	1.95V	28 ns	17 ns	117 ns	35 ns
IRGB4064DPBF	TO-220	10.0A	2.00V	27 ns	16 ns	98 ns	33 ns
IRGB4056DPBF	TO-220	12.0A	1.97V	30 ns	18 ns	102 ns	41 ns
IRGB4061DPBF	TO-220	18.0A	2.15V	40 ns	25 ns	120 ns	40 ns
IRGP4062DPBF	TO-247	24.0A	2.04V	40 ns	24 ns	125 ns	39 ns
IRGB4062DPBF	TO-220	24.0A	2.04V	40 ns	24 ns	125 ns	39 ns
IRGP4063DPBF	TO-247	48.0A	2.10V	55 ns	45 ns	165 ns	45 ns

600V Half Bridge Gate Driver ICs*

Part Number	Package Type	I_{O+}	I_{O-}
IRS2113STRPBF	SOIC	2.5 A	2.5 A
IR2114SSTRPBF	SOIC	2.0 A	3.0 A

600V Low Uce(sat) IGBTs for less than 1kHz*

Part Number	Package Type	I_C at 100°C	$V_{CE(on)}$ at Rated Current	Q_g	$R_{th(j-c)}$
IRG4PC50SDPBF	TO-247	41A	1.28V	180 nC	0.64 $^{\circ}\text{C}/\text{W}$
IRG4PC40SPBF	TO-247	31A	1.32V	100 nC	0.77 $^{\circ}\text{C}/\text{W}$
IRG4BC30SPBF	TO-220	18A	1.4V	50 nC	1.2 $^{\circ}\text{C}/\text{W}$
IRG4BC20SDPBF	TO-220	10A	1.4V	27 nC	2.1 $^{\circ}\text{C}/\text{W}$

*Also available in 1200V

For more information, please visit us at www.irf.com or www.irf.com.cn.

Features

- Optimized 600V trench IGBTs, 600V standard speed IGBTs and HVIC for the inverter side
- 30% reduction in power loss using trench and standard speed planar IGBT combination versus ultrafast speed planar IGBT alone

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